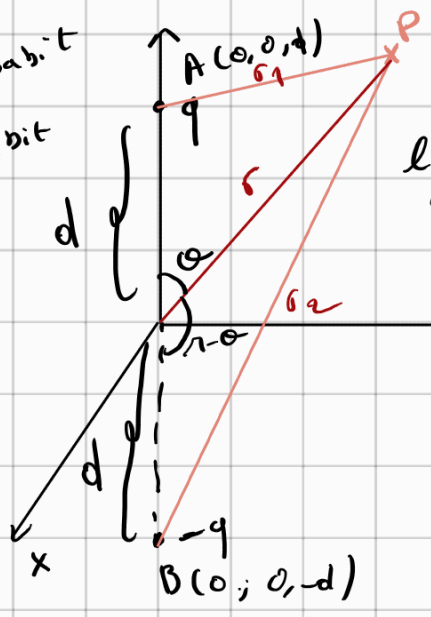
Dipol (Elektriksel)

$\lim_{d\to 0}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)=\frac{1}{r}-\frac{1}{r}=0$

$p=q\cdot d$ (C_m) = sabit

$\lim_{\substack{q\to\infty\\ d\to 0}} qd=p=$ sabit



$$V=\frac{q}{4\pi\varepsilon_0}\frac{1}{r_1}-\frac{q}{4\pi\varepsilon_0}\frac{1}{r_2}$$

$$\lim_{\substack{q\to\infty\\ d\to 0}} V = V_{dipol} = \frac{1}{4\pi\varepsilon_0}\lim_{\substack{q\to\infty\\ d\to\infty}}\frac{qd}{d}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\lim_{\substack{q\to\infty\\ d\to 0}} qd \lim_{\substack{q\to\infty\\ d\to 0}}\frac{1}{d}\left(\frac{1}{r_1}-\frac{1}{r_2}\right)$$

$$V_{dipol}=\frac{p}{4\pi\varepsilon_0}\cdot\lim_{d\to 0}\frac{\left(\frac{1}{r_1}-\frac{1}{r_2}\right)}{d}$$

$\triangle OAP$ $\quad r_1=\sqrt{r^2+d^2-2rd\cos\theta}$

$\triangle OBP$ $\quad r_2=\sqrt{r^2+d^2-2rd\cos(\pi-\theta)}=\sqrt{r^2+d^2+2rd\cos\theta}$

$$V_{dipol}=\frac{p}{4\pi\varepsilon_0}\lim_{d\to 0}\frac{\left(\frac{1}{r_1}-\frac{1}{r_2}\right)}{d}$$

$$=\frac{p}{4\pi\varepsilon_0}\lim_{d\to 0}\frac{\frac{1}{\sqrt{r^2+d^2-2rd\cos\theta}}-\frac{1}{\sqrt{r^2+d^2+2rd\cos\theta}}}{d}$$

$$=\frac{p}{4\pi\varepsilon_0}\lim_{d\to 0}\frac{d}{d(d)}\left[(r^2+d^2-2rd\cos\theta)^{-1/2}+(r^2+d^2+2rd\cos\theta)^{-1/2}\right]$$

$$V_{dipol}=\frac{p}{4\pi\varepsilon_0}\lim_{d\to 0}\left[-\frac{1}{2}\frac{(2d-2r\cos\theta)}{(r^2+d^2-2rd\cos\theta)^{3/2}}+\frac{1}{2}\frac{(2d+2r\cos\theta)}{(r^2+d^2+2rd\cos\theta)^{3/2}}\right]$$

$$V_{dipol}=\frac{-p}{4\pi\varepsilon_0}\lim_{d\to 0}\left[\frac{d-r\cos\theta}{(r^2+d^2-2rd\cos\theta)^{3/2}}-\frac{d+r\cos\theta}{(r^2+d^2-2rd\cos\theta)^{3/2}}\right]$$

$2p=2qd$
$=q(2d)$

$$=-\frac{p}{4\pi\varepsilon_0}\left[-\frac{r\cos\theta}{r^3}-\frac{r\cos\theta}{r^3}\right]=\frac{p\cos\theta}{2\pi\varepsilon_0 r^2}$$

p → dipolun moment yoğunluğu

$$V=\frac{2p\cos\theta}{4\pi\varepsilon_0 r^2}$$

kitaptaki

z ekseni boyunca oluşturulmuş $p=qd$ moment yoğunluk bir noktasal dipolun potansiyeli:

p = dipol moment yoğunluğu $[p] = C\,m$

$[V_{dipol}] = V$

$$\text{grad } f = \frac{\partial f}{\partial r}\vec{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\vec{e}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}\vec{e}_\phi \qquad f(r,\theta,\phi)$$

$$E = -\text{grad } V = -\frac{\partial V_{dipol}}{\partial r}\vec{e}_r - \frac{1}{r}\frac{\partial V_{dipol}}{\partial \theta}\vec{e}_\theta - \frac{1}{r\sin\theta}\frac{\partial V_{dipol}}{\partial \phi}\vec{e}_\phi$$

$$= \frac{p\cos\theta}{4\pi\varepsilon_0 r^3}\vec{e}_r + \frac{1}{r}\frac{p}{4\pi\varepsilon_0 r^2}\sin\theta\,\vec{e}_\theta$$

kitaptakiler →

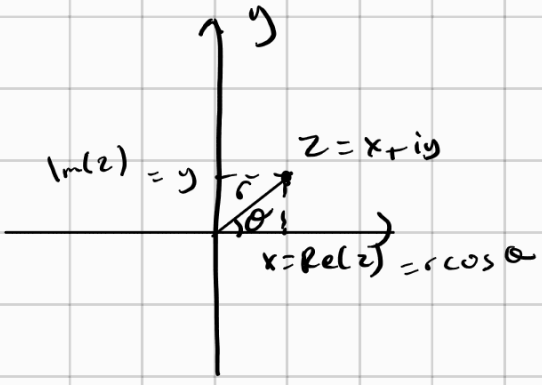
$$= \frac{p}{4\pi\varepsilon_0}\left[2\cos\theta\,\vec{e}_r + \sin\theta\,\vec{e}_\theta\right]$$

Yada en baştan $\vec{E} = \frac{q}{4\pi\varepsilon_0}\frac{\vec{e}_{r_1}}{r_1^2} - \frac{q}{4\pi\varepsilon_0}\frac{1}{r_2}\vec{e}_{r_2}$

ama çokdaha zoooooor!!

$\star\ \text{rot}\,\vec{E}_{dipol} \equiv 0$

Hadi kompleks sayılardan bakışedelim



$z = Re(z) + i\,Im(z)$

$r = \sqrt{x^2+y^2} = |z|$

$z = r\cos\theta + i r\sin\theta$

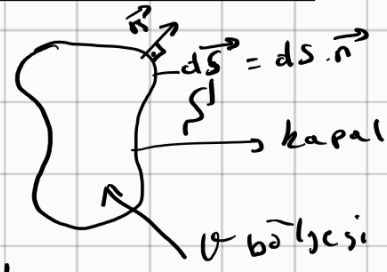
$= r[\cos\theta + i\sin\theta]$

$= |z|[\cos(\arg z) + i\sin(\arg z)]$

Gauss Teoremi

$$\vec{A} = A_x \vec{e}_x + A_y \vec{e}_y + A_z \vec{e}_z$$

$A_x, A_y, A_z (x, y, z)$



v S tarafından kuşatılan bölge

Gauss - Ostrogradski teoremi

$$\oint_S \vec{A} \cdot d\vec{s} = \int_v \operatorname{div} \vec{A} \cdot dv$$

$$\iint_S \vec{A} \cdot d\vec{s} = \iiint_v \operatorname{div} \vec{A} \cdot dv$$